Experiment No : 3

Title :

Implement Control Structures and Form Validation; Create a Grading System Based on User-Entered Marks

Software/Tools Required :

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome, Microsoft Edge, Mozilla Firefox, etc.)
- Basic knowledge of HTML, CSS, and JavaScript

Theory :

JavaScript is a powerful client-side scripting language used to make web pages interactive and dynamic. It provides various **control structures**, **loops**, and **form validation techniques** that allow developers to process user input, make logical decisions, and display dynamic results without reloading the webpage. These features are essential for developing responsive and user-friendly web applications.

1. if-else Statement

The **if-else** statement is a decision-making control structure that executes different blocks of code based on whether a specified condition evaluates to **true** or **false**. It is widely used in applications where different outcomes depend on user input or program conditions.

Applications:

- Assigning grades based on percentage.
- Checking whether marks are within the valid range.
- Displaying appropriate error messages.
- Verifying login credentials.

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The **switch** statement is used when multiple conditions need to be evaluated against a single expression. Instead of writing multiple if-else statements, a switch statement improves readability and simplifies the code by selecting the matching case.

Advantages:

- Makes code easier to understand.
- Reduces the complexity of multiple condition checks.
- Suitable for menu-driven programs and option selection.

3. Loops

Loops are used to execute a block of code repeatedly until a specified condition is satisfied. They reduce code duplication and improve efficiency.

a) for Loop

- Executes code for a predetermined number of iterations.
- Commonly used when the number of repetitions is already known.

Example Uses:

- Displaying a list of students.
- Processing multiple subject marks.
- Generating tables.

b) while Loop

- Executes the code as long as the specified condition remains true.
- Best suited when the number of iterations is unknown beforehand.

c) do-while Loop

- Executes the code at least once before checking the condition.
- Useful when at least one execution is required regardless of the condition.

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Form validation is the process of verifying user input before it is processed by the application. Validation improves data accuracy, prevents invalid entries, and enhances the overall user experience.

A well-designed validation system checks:

- Whether required fields are empty.
- Whether marks are numeric.
- Whether marks fall within the valid range (0–100).
- Whether invalid characters have been entered.
- Whether all mandatory information has been provided.

Proper validation prevents incorrect calculations and ensures reliable program output.

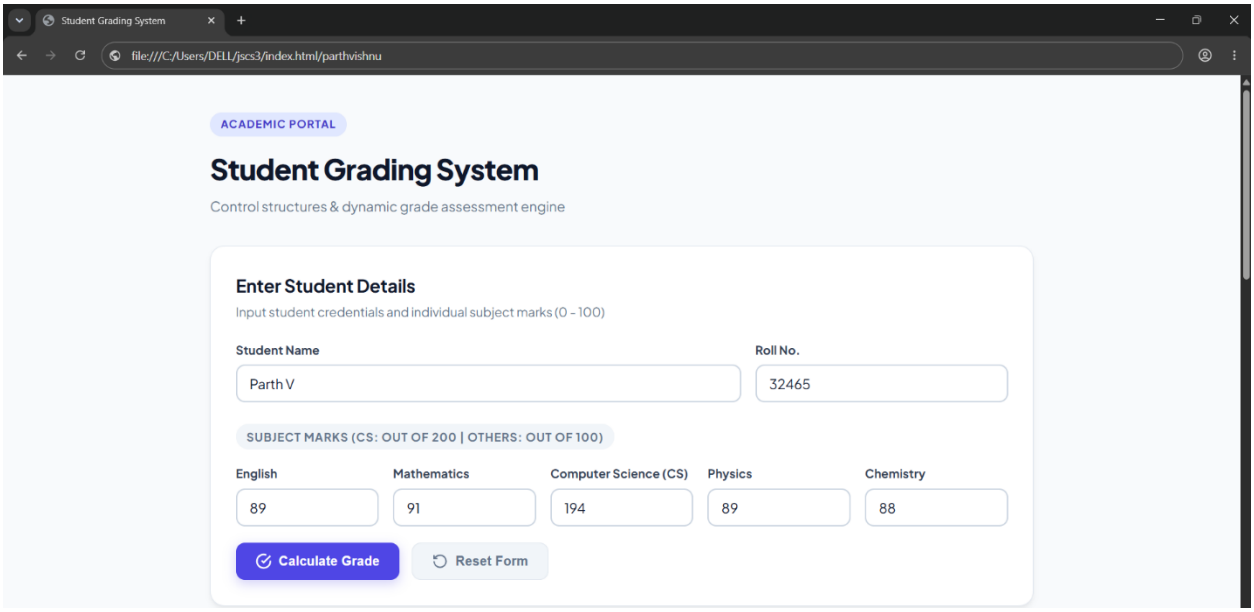
Aim of the CASE STUDY

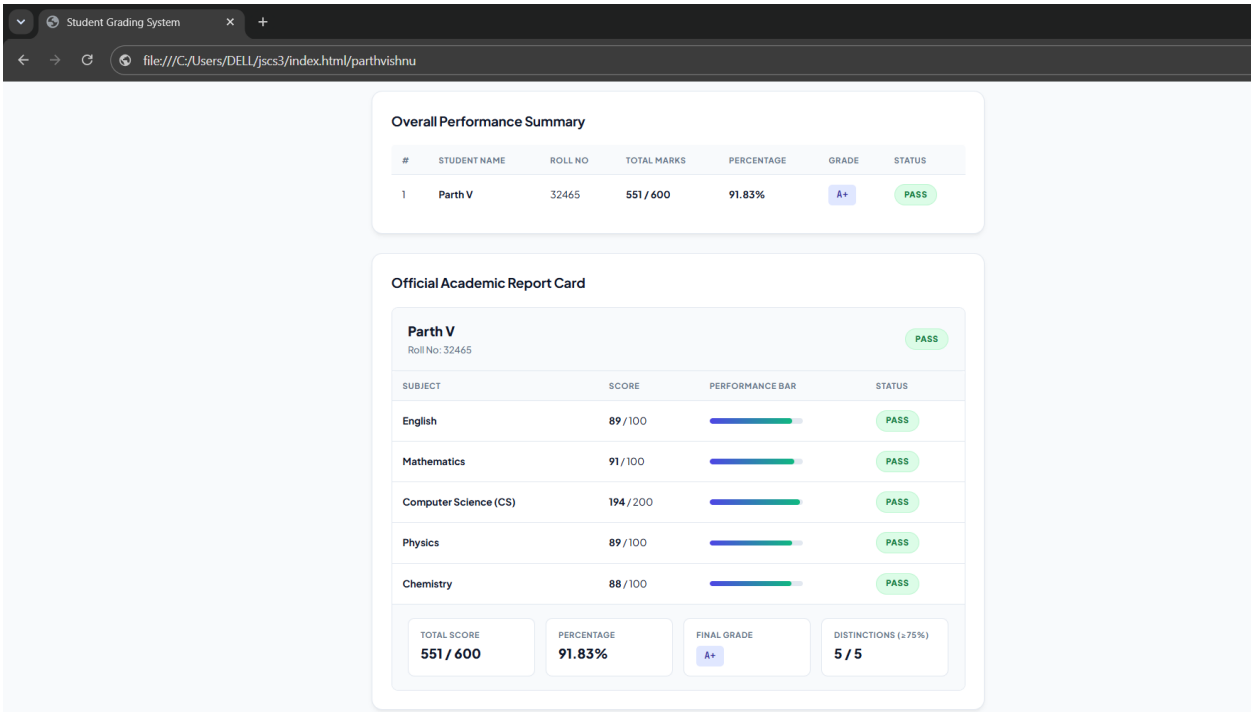
The **Student Grading System** is a JavaScript-based web application that demonstrates the use of control structures and form validation. The application accepts student information, validates the entered data, calculates academic performance, and displays the final result dynamically.

Working of the System

1. The user enters the student's name, roll number, and marks.
2. The program validates every input field.
3. If any field is empty or contains invalid data, an error message is displayed.
4. If all inputs are valid, the program:
    - Calculates the total marks.
    - Computes the percentage.
    - Determines the grade using **if-else** or **switch** statements.
5. The final result is displayed on the webpage without refreshing it.

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Output :

After running the program in a web browser, a form is displayed where the user enters the student's details and marks obtained in different subjects. Once the **Submit** or **Calculate** button is clicked, the JavaScript program first validates all the entered information.

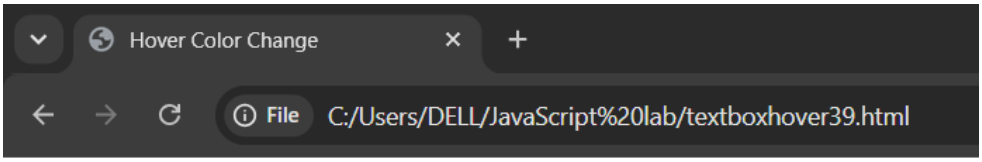
displays an appropriate validation message and prevents further processing until the errors are corrected.

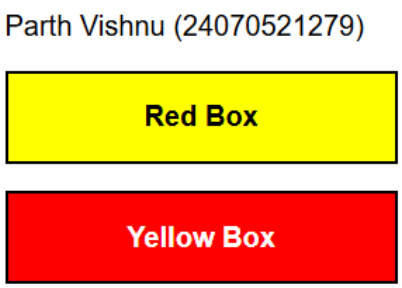
If all the entered values are valid, the application performs the following operations:

- Calculates the total marks obtained by the student.
- Computes the overall percentage.
- Determines the grade according to the predefined grading criteria.
- Displays the student's details, total marks, percentage, grade, and pass/fail status dynamically on the webpage.

The output is generated instantly without reloading the page, demonstrating the dynamic nature of JavaScript.

Additional assignment :





Conclusion :

This experiment demonstrated the use of JavaScript control structures, loops, and form validation by creating a Student Grading System. The application successfully validates inputs, calculates marks and percentage, assigns grades using conditional statements, and displays results dynamically. It provided practical understanding of JavaScript decision-making and form handling for developing interactive web applications. Also a short assignment of box colour swapping on mouse hover(event).